

ROQUEMAURE, QC, Canada

WMO: 717361

Lat: 48.630N

Lon: 79.460W

Elev: 884

StdP: 14.23

Time Zone: -5.00 (NAE)

Period: 93-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.0	-27.7	-42.5	0.5	-33.5	-36.3	0.7	-27.2	19.8	12.0	18.2	9.1	2.6	180	0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	83.4	68.9	80.1	66.9	76.9	64.8	71.2	79.3	69.2	76.6	67.2	73.9	7.5	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.6	108.5	75.3	66.6	101.2	73.2	64.6	94.2	71.7	35.7	79.8	33.9	76.1	32.3	73.7	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.7	16.5	14.5	DB	-42.9	88.5	4.9	3.5	-46.4	91.0	-49.3	93.0	-52.0	95.0	-55.6	97.5	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.5	-0.4	3.5	17.2	31.9	47.1	59.2	62.6	60.1	52.3	40.2	25.9	12.3
	DBStd	24.25	15.17	14.73	14.41	10.18	8.48	8.41	6.06	6.60	8.14	8.20	10.88	15.01
	HDD50	6893	1563	1302	1018	547	162	17	0	3	63	326	723	1169
	HDD65	11264	2028	1722	1483	994	558	212	119	183	390	769	1172	1634
	CDD50	1231	0	0	0	3	72	294	392	316	132	22	0	0
	CDD65	127	0	0	0	0	2	39	46	31	9	0	0	0
	CDH74	1355	0	0	0	0	77	464	424	291	99	0	0	0
	CDH80	297	0	0	0	0	11	119	88	54	25	0	0	0
Wind	WSAvg	6.4	6.1	6.7	6.5	6.4	6.8	5.8	5.8	5.6	6.3	7.7	7.2	6.5
Precipitation	PrecAvg	31.5	2.0	1.4	1.7	2.0	2.6	2.9	3.8	3.4	3.7	3.3	2.6	2.1
	PrecMax	38.8	4.5	3.0	3.2	3.9	4.8	5.1	7.0	5.3	7.2	5.6	4.9	3.7
	PrecMin	25.9	0.6	0.2	0.2	0.7	0.9	1.7	1.9	1.1	1.6	1.3	1.5	1.1
	PrecStd	3.0	0.9	0.7	0.6	0.8	1.0	1.0	1.2	1.0	1.1	1.1	0.8	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.4	44.1	53.6	65.1	81.8	88.2	87.7	85.5	83.5	69.5	56.9	40.1
		MCWB	36.6	41.2	44.8	55.3	63.6	71.2	73.2	70.0	66.5	60.5	51.0	36.2
	2%	DB	32.3	38.0	46.9	58.1	75.4	83.4	81.9	80.8	75.8	63.2	50.6	36.8
		MCWB	31.3	34.5	41.1	46.9	57.9	68.3	69.4	67.6	65.5	56.8	48.9	35.4
	5%	DB	28.3	33.0	41.9	52.4	69.8	79.5	78.7	77.3	70.2	58.3	44.9	35.1
		MCWB	27.3	31.5	36.7	42.2	55.2	65.0	67.2	66.5	62.7	54.1	42.5	33.9
	10%	DB	24.0	28.1	37.7	47.8	64.5	75.5	75.4	73.3	66.8	53.8	39.7	32.8
		MCWB	22.8	26.1	33.8	40.5	51.2	63.0	65.3	64.5	60.6	50.0	37.5	31.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.6	42.4	46.1	56.3	63.0	73.6	75.2	72.6	69.1	62.1	51.8	37.6
		MCDB	37.2	44.1	52.5	62.9	75.8	84.6	84.3	79.7	76.3	66.9	53.8	38.4
	2%	WB	30.9	34.6	41.2	48.7	58.8	70.3	71.8	70.2	66.9	58.6	48.4	35.6
		MCDB	31.5	37.2	45.8	51.8	67.6	80.6	78.4	77.7	73.6	62.6	51.1	36.4
	5%	WB	26.6	31.1	36.7	44.0	55.5	68.2	69.2	68.1	64.3	54.7	43.4	34.0
		MCDB	27.8	32.9	40.3	50.1	63.9	75.8	75.2	75.4	70.5	58.2	45.7	35.0
	10%	WB	22.3	25.2	33.8	40.5	52.8	65.5	66.6	65.9	60.8	50.4	38.1	31.9
		MCDB	23.5	26.8	36.6	46.8	60.4	72.3	72.7	72.2	65.0	53.7	40.1	33.0
Mean Daily Temperature Range	5% DB	MDBR	22.9	27.2	24.2	21.7	23.1	23.5	20.1	21.2	19.4	14.9	13.8	17.4
		MCDBR	21.2	23.0	23.4	28.7	32.5	28.3	25.0	25.8	25.3	21.3	17.1	11.3
		MCWBR	20.2	19.5	18.1	16.8	18.1	16.1	14.0	14.7	15.7	15.7	14.8	10.5
	5% WB	MCDBR	21.3	21.5	20.5	22.3	26.1	23.5	21.8	21.7	23.0	19.1	16.1	10.5
		MCWBR	21.2	19.5	16.2	16.6	15.6	15.0	13.3	13.4	15.2	14.9	14.1	10.3
Clear-Sky Solar Irradiance	taub	0.261	0.278	0.311	0.351	0.359	0.377	0.393	0.381	0.364	0.320	0.293	0.255	
	taud	2.283	2.241	2.220	2.293	2.398	2.370	2.360	2.386	2.417	2.529	2.460	2.351	
	Ebn at Noon	257	278	285	285	286	280	273	272	266	263	245	245	
	Edh at Noon	23	30	37	38	36	37	37	35	30	23	20	19	
All-Sky Solar Radiation	RadAvg	393	744	1185	1462	1646	1803	1706	1431	1068	624	322	260	
	RadStd	63	103	118	118	121	151	112	148	95	96	87	37	37

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page